20. OTHER INCOME AND EXPENSES, NET

Other income and expenses, net consisted of the following:

	Year ended December 31, 2019	Year ended December 31, 2018	Year ended December 31, 2017
Research and development funding	132	52	65
Phase-out and start-up costs	(38)	(1)	(8)
Exchange gain (loss), net	—	4	4
Patent costs	(1)	(8)	(9)
Gain on sale of businesses and non-current assets	7	8	4
Other, net	3	(2)	(1)
Total	**103**	**53**	**55**

The Company receives significant public funding from governmental agencies in several jurisdictions. Public funding for research and development is recognized ratably as the related costs are incurred once the agreement with the respective governmental agency has been signed and all applicable conditions have been met. R&D funding received in the year ended December 31, 2017 from the Nano2017 program with the French government is subject to a financial return in the year 2024 and depends on the future cumulative sales of a certain product group from 2019 to 2024. As such, an accrual amounting to $47 million was recorded as of December 31, 2019 compared to $42 million as of December 31, 2018.

Phase-out costs are costs incurred during the closing stage of a Company's manufacturing facility. They are treated in the same manner as start-up costs. Start-up costs represent costs incurred in the start-up and testing of the Company's new manufacturing facilities, before reaching the earlier of a minimum level of production or six months after the fabrication line's quality certification.

Exchange gains and losses, net represent the portion of exchange rate changes on transactions denominated in currencies other than an entity's functional currency and the changes in fair value of trading derivative instruments which are not designated as hedge and which have a cash flow effect related to operating transactions, as described in Note 27.

Patent costs include legal and attorney fees and payment for claims, patent pre-litigation consultancy and legal fees. They are reported net of settlements, if any, which primarily include reimbursements of prior patent litigation costs.

In 2019, gain on sale of businesses and non-current assets was related to the sale of one of our non-strategic assets. In 2018, it was related to the sale of one of the Company's non-strategic investments while in 2017, it was related to the sale of assets.

21. IMPAIRMENT, RESTRUCTURING CHARGES AND OTHER RELATED CLOSURE COSTS

Impairment, restructuring charges and other related closure costs incurred in 2019, 2018 and 2017 are summarized as follows:

Year ended December 31, 2019	Impairment	Restructuring charges	Other related closure costs	Total impairment, restructuring charges and other related closure costs
Set-top Box restructuring plan	—	2	—	2
Long-lived asset impairment charge	(7)	—	—	(7)
Total	**(7)**	**2**	**—**	**(5)**